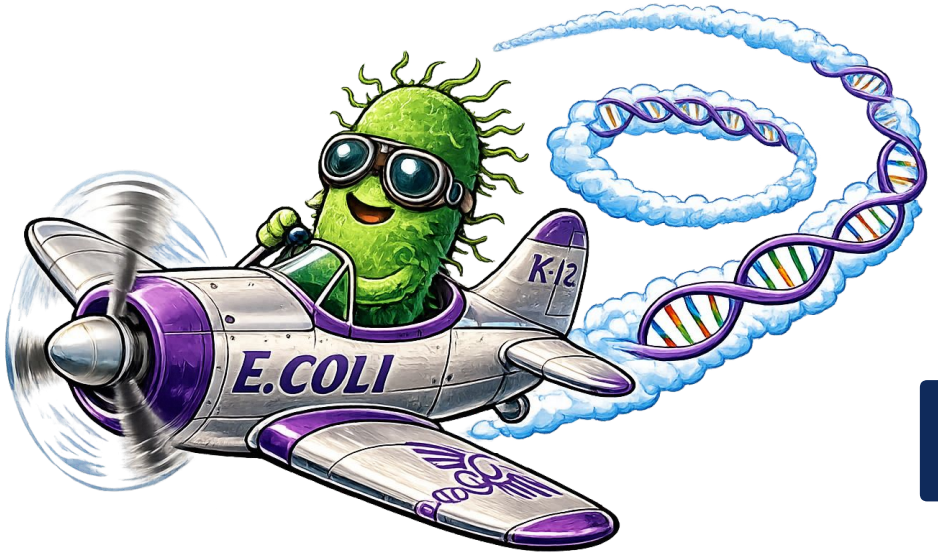


BioPilot

A Copilot for End-to-End Generative Design of Synthetic Biology Circuits

Charles Van Hook · Joshua Kome · Michael Wallace

EC/BE 552 · Computational Synthetic Biology for Engineers
Prof. Douglas Densmore · Spring 2026

Goal & Motivation

Why a copilot for synthetic biology circuits

- Problem — synthetic biology design today is fragmented across parts registries, combinatorial-design tools, and sequence-generation models; no single tool spans prompt → buildable DNA.
- Goal — give a wet-lab researcher one natural-language entry point that (a) finds the right real parts, (b) assembles them into a combinatorial design space, and (c) lets a foundation model propose alternative sequences per slot.
- Concrete use case — a biosensor lab: "Design me an arsenic biosensor that glows green." BioPilot returns a GOLDBAR design space built from iGEM / GenBank / UniProt parts, per-design DNA sequences, and Evo 2-proposed alternates for the RBS and terminator.
- Synthetic biology connection — this IS computational synthetic biology: we compose characterized parts into circuits, express them as combinatorial design rules, and score / regenerate sequences with a DNA language model.
- Why it matters — shortens the "idea → first-draft plasmid map" loop from days of manual literature + registry browsing to a single prompt.

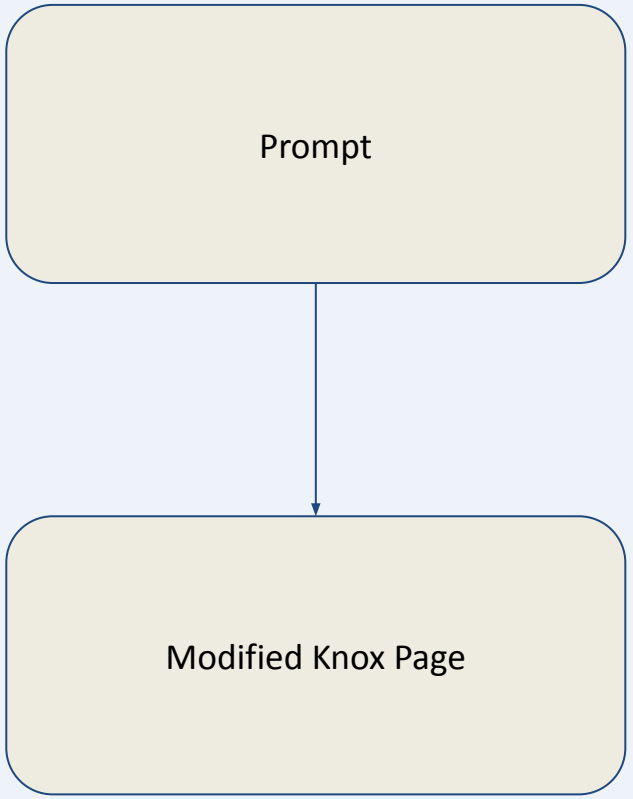
presenter: Mike

Short Functionality

What BioPilot does, in one slide

- You type a plain-English design request in a chat window
- An LLM agent reads intent, picks real parts from a curated library of 7,000+ biological parts, and assembles them into a combinatorial design space.
- The design space is browsable as a graph — every enumerated design has complete, buildable DNA.
- For any slot in any design, EVO 2 can propose an alternative sequence conditioned on its actual upstream context.

User sees:



presenter: Charles

Short Specification

What BioPilot had to do — constraints and numbers

- Single entry point — one natural-language prompt, no structured schema or GUI forms.
- No hallucinated DNA — every part in every design traces back to a real registry entry (iGEM, GenBank, UniProt, Addgene).
- Combinatorial, not single-shot — output is a design SPACE of enumerable alternatives, not one proposed circuit.
- Latency — under a minute from prompt to browsable design space.
- Standards-compliant output — GOLDBAR combinatorial spec + SBOL so downstream CAD tools can consume it.

7,264
real biological parts in Qdrant

4
LLM providers supported (Gemini default)

8
built-in circuit topology templates

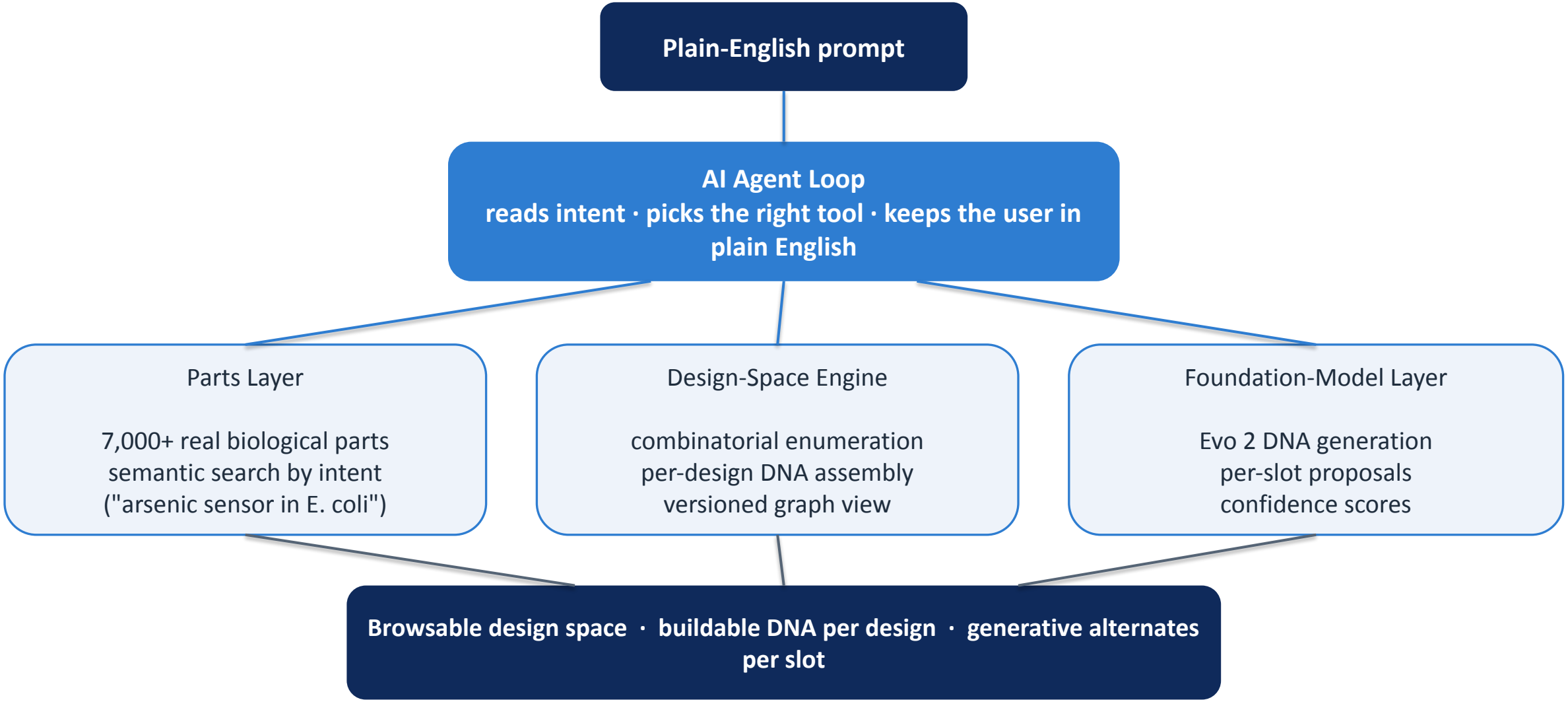
40 B
parameters in Evo 2 (NVIDIA NIM)

~45 s
measured NL-prompt to imported design

104
tests passing in scorer subsystem

System Architecture

Capability layers — what each block does for the user



The agent never touches DNA directly — it only decides WHICH capability to call. Parts come from the parts layer; combinatorial structure and sequences come from the design-space engine; generative alternates come from the foundation-model layer.

presenter: Mike

Subsystem Detail

What each capability layer enables

Parts Layer

owner: Charles

Turns intent into a list of buildable real parts

- 7,000+ parts and designs curated from iGEM, GenBank, UniProt, Addgene, Cello UCF — all normalized to one schema.
- Semantic search: ranks parts by description, role, and organism.
- Assembly engine fills each slot with the best-matching real part.
- Exposed as a tool server so any LLM client can consume the same capability.

Design-Space Engine

owner: Mike

Turns a prompt into a combinatorial design space

- Agent-driven chat orchestrates the entire pipeline end-to-end.
- Remote parts tools appear alongside native design tools in one registry.
- GOLDBAR combinatorial grammar → versioned graph of enumerated designs.
- Per-design DNA assembly with color-coded slot boundaries.
- One-click regenerate-this-slot wired to the foundation-model layer.
- Standards-compliant SBOL export for downstream CAD tools.

Foundation-Model Layer

owner: Josh

Brings a DNA foundation model into the designer's hands

- Evo 2 integration — SOTA DNA language model (1B for dev, 40B hosted for production).
- Parses Cello UCF libraries into typed part schemas.
- Multi-objective scorer: GC content, folding energy, ribosome affinity.
- CircuitValidator checks topology and sequence sanity.
- Ablation study: 4 conditioning modes on the CELLO circuit data
- Correlation study: Evo2 scoring vs RBS ground-truth

Clear Example 1 — Prompt to Design Space

The core capability: a plain-English request becomes a combinatorial design space with real DNA

User, in chat:
"Design an arsenic biosensor in E. coli that produces GFP."

System allows (in ~45 s):
Enumerated designs · full DNA + graph view

1. Intent

Plain English in. No schema, no form, no part IDs. The user describes the goal.

2. Plan

LLM agent reads intent → decides which capability to call (here: parts search for a biosensor topology in E. coli).

3. Real parts

Parts layer searches 7,000+ curated entries and returns ranked candidates per role: promoter (Pars), RBS, CDS (GFP), terminator.

4. Design space

Design-space engine emits a GOLDBAR combinatorial spec, imports it as a versioned graph, and enumerates all compatible designs.

5. Buildable DNA

Per-design DNA is assembled and rendered as color-coded slots — ready to order, or hand off to the generative layer (example 2).

Foundational Capability:

- Before BioPilot: a designer manually browses parts registries, picks parts by hand, writes GOLDBAR by hand, imports by hand — lots of work.
- With BioPilot: one sentence produces a combinatorial design space with real, provenance-tracked DNA — no hallucinated sequences, every part traceable.
- This is the backbone that example 2 (Evo 2 generative fill) plugs into — without a grounded design space to sit on top of, the foundation model has nothing to condition on.

presenter: Mike

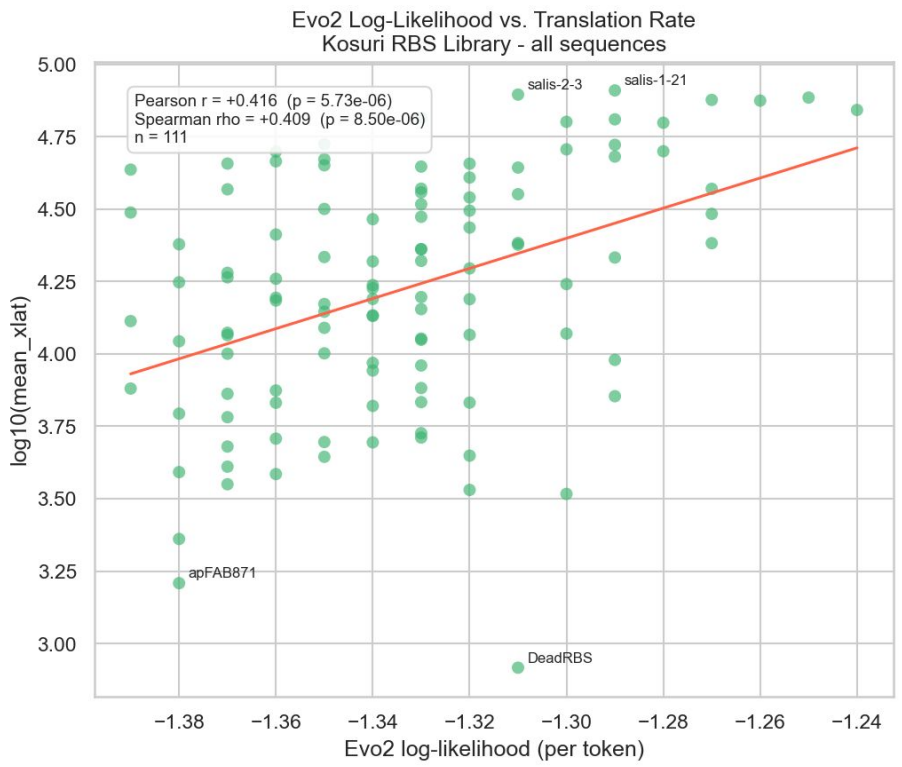
Evo 2 Enables Generative Fill

Integration of DNA foundation models easily create designs with *de novo* generated parts

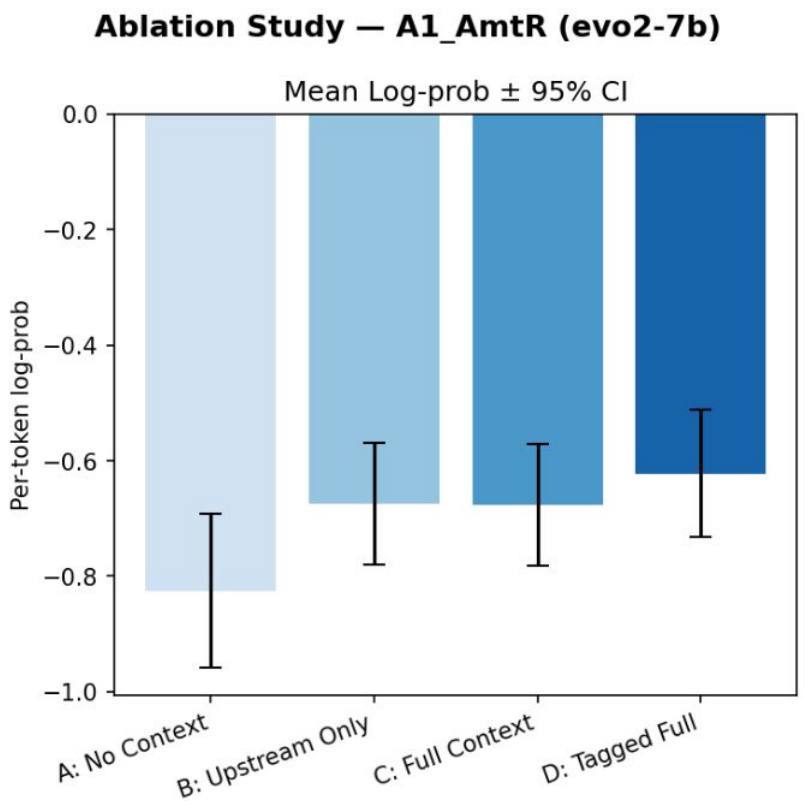
Evo 2 is a DNA foundation model trained on 9 trillion bases over all kingdoms of life

2 validation experiments to demonstrate use cases for synthetic circuit design

1. Validate against ground-truth RBS data



2. Ablation study with synthetic circuit context



presenter: Josh

Clear Example 2 — Evo 2 Generative Fill

Integration of DNA foundation models easily create designs with *de novo* generated parts

Promoter (Pars)	RBS (BBa_B0034)	CDS (GFP)	Terminator (B0015)

Regenerate with Evo 2

1. Click 'View'

Knox opens the sequence viewer modal with a color-coded slot strip.

2. Rebuild 5' context

SequenceTools concatenates DNA from all prior slots (promoter + RBS).

3. Call Evo 2 40B

POST to NVIDIA NIM with num_tokens = original slot length, top_k=1.

4. Receive proposal

Evo 2 returns new DNA + mean sampled probability.

5. Diff view

Original vs Evo 2 side-by-side; user keeps, rejects, or re-rolls.

Why this is the thing we're most proud of

- The foundation model is an active designer, not a passive ranker — it proposes replacement DNA rather than scoring an enumeration.
- Conditioning is real: Evo 2 sees the actual upstream 5' DNA of THIS design, not a generic prompt.
- One browser click bridges three worlds — a Gemini agent's curated pick, a GOLDBAR-enumerated design space, and a 40B DNA language model — with a quantitative mean sampled probability for the user to compare.

Successes

What worked, and why

- Shipped the end-to-end capability: plain-English prompt → combinatorial design space → buildable DNA.
- 7,000+ real biological parts indexed and semantically searchable — every design traceable to registry entries.
- Agent reliably decomposes prompts across diverse topologies — biosensor, logic-NOT, and metabolic-pathway requests all produce correct designs.
  - Anti-hallucination discipline held: zero invented sequences across the demo suite.
  - Host-organism coherence rules eliminated chimeric E. coli / B. subtilis / mammalian mash-ups from the design space.
- Evo 2 Generative Fill allows users to generate *de novo* parts in real-time with confidence scores, plugging directly into Knox design-space view.
- All three subsystems talk through typed data contracts — changes in one layer don't break the others.

presenter: Mike

Limitations & Later Reach Goals

What we'd do next — and what we'd do differently

Current limitations

- Evo 2 Generative Fill uses upstream-only (autoregressive) context — full bidirectional infill is a later reach goal.
- Protein CDS entries (UniProt) have no DNA representation, so regenerate is disabled on slots downstream of a protein part
- Large designs get context-truncated for latency on the hosted foundation model; global conditioning drops off past a few kb.
- BioPilot is a local developer tool today — not a hosted service.
- Smaller curated parts database for demo

What we'd do differently — reach goals

- Rank enumerated designs with a foundation model and use back as a second-pass score across enumerated Evo 2 part proposals.
- Bidirectional infill — swap in a masked DNA language model so any slot can be regenerated with full upstream AND downstream context.
- Surface the multi-objective biophysical scorer (GC, folding, ribosome affinity) right next to each part and each generative alternate.
- Host BioPilot as a service — move from a local developer tool to a URL anyone can visit and use.

presenter: All

Attribution of Work

Charles
Parts Engine

MCPGeneBank

- NL-to-circuit pipeline design
- 7,264-part Qdrant vector index
- API and webs scrape data collection
- BioPart schema + part normalization
- LLM planner + CircuitSpec
- Assembly engine & topology templates
- MCP SSE tool server (FastAPI)

Mike
Integration & Agent

Design-space engine + end-to-end glue

- Agent-driven chat + unified tool registry
- Cross-layer bridge (parts ⇄ design space)
- GOLDBAR auto-generation & import
- Sequence assembly + Fetch Sequences UI
- Evo 2 Generative Fill tooling + diff view
- End-to-end demo orchestration

Josh
Foundation Model Layer

generative-syn-bio / Evo 2

- Evo 2 integration (1B / 7B / 40B NIM)
- Cello UCF parser + PartSpec schema
- Multi-objective scorer (GC, MFE, SD)
- CircuitValidator
- Ablation study on Mutalik RBS library
- 104-test scorer test suite

Each pillar ships independently; the integration layer guarantees they compose. All three members contributed to design reviews, demo rehearsal, and documentation.

Live Demo & Q&A

Give us a request!

Chat window → real parts → design space → DNA

Example Prompts:

Proven biosensors:

1. Design an arsenic biosensor that produces GFP
2. Design an arsenic-sensing NOT gate that produces RFP

Inducible expression:

3. Design a circuit where arabinose induces GFP expression
4. Design a circuit where tetracycline turns off GFP expression

Classic cell-cell & memory

5. Design a genetic toggle switch with LacI and TetR, one state glows red the other glows green
6. Design a LuxI / LuxR quorum-sensing sender with GFP reporter

Reporter library / tuning

7. Make five constitutive promoter variants driving GFP at different strengths
8. Build an IPTG-inducible mCherry reporter

presenter: Mike

Live Demo & Q&A

"Design an arsenic biosensor in E. coli that produces GFP."

Chat window → real parts → design space → DNA

Team Information:

Repository: github.com/mswallac/BE552_Project

Team: Charles Van Hook · Joshua Kome · Michael Wallace

Course: EC/BE552 — Computational Synthetic Biology for Engineers — Spring 2026

Questions?

Backup Demo

KNOX

Space IDs

arsenic_biosensor_v1

blue_green_oscillator

GROUP

AI Chat

AI: Welcome to Knox! I am your AI assistant. I can help you with questions about GOLDBAR, combinatorial design, and how to use Knox.

You can ask me to retrieve information about design spaces and groups.

I can also apply operators to design spaces for you.

Feel free to ask me anything related to these topics!

You: make me a blue-green oscillator

AI: Imported design space **blue_green_oscillator**
GOLDBAR: `(s0_0_promoter . s0_1_ribosome_entry_site . s0_2_CDS . s0_3_terminator) . (s1_0_promoter . s1_1_ribosome_entry_site . s1_2_CDS . s1_3_terminator) . (s2_0_promoter . s2_1_ribosome_entry_site . s2_2_CDS . s2_3_terminator) . (s3_0_promoter . s3_1_ribosome_entry_site . s3_2_CDS . s3_3_terminator) . (s4_0_promoter . s4_1_ribosome_entry_site . s4_2_CDS . s4_3_terminator)`
This design is a three-component repressilator aiming for blue-green oscillation. TU1 expresses cI repressor under the control of TetR-repressible promoters (pButR/pTarA

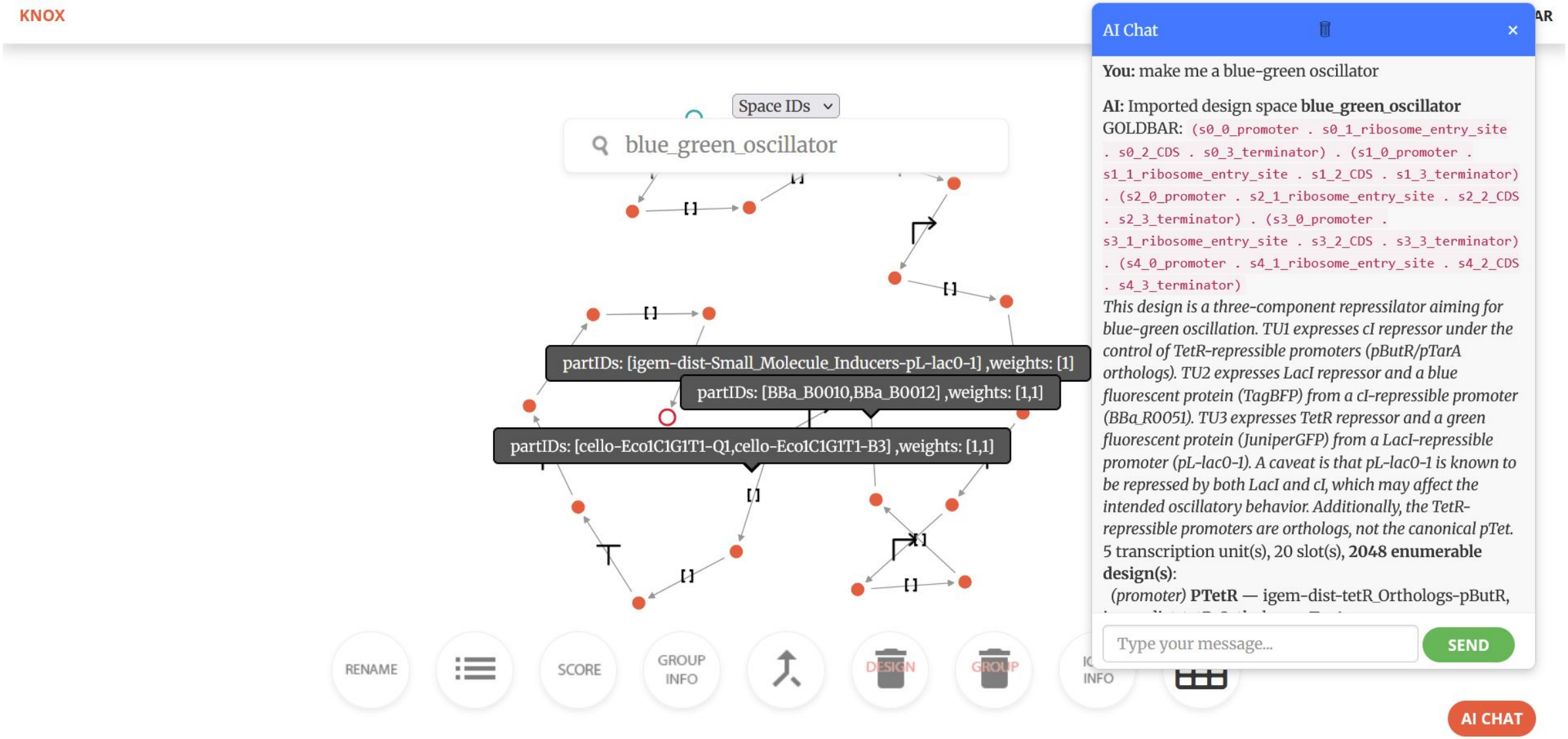
Type your message...

SEND

AI CHAT

presenter: Mike

Backup Demo



presenter: Mike

Backup Demo

Enumerated Designs

Export All to CSV | Fetch Sequences

Assembled DNA sequences for 3 design(s) from
blue_green_oscillator
(12 unique parts, 1 batch MCP call)

>blue_green_oscillator design 0 | 5082 bp View

```
ttcGTGTCACTTTGACAGCAGTGTCACtcctaggtataatgctagcGTAAGCCA
TGCCATTGGCTTTTGATAGAGGATAACTACTAGATGGCCTTGGTTGACGGTTTT
CTTGAGCTGGAACGCTCAAGTGGAAAATTGGAGTGGAGCGCCATCCTCCAGAAG
ATGGCGAGCGACCTTGGATTCTCGAAGATCCTGTTCGGCCTGTTGCCTAAGGAC
AGCCAGGACTACGAGAACGCCTTCATCGTCGGCAACTACCCGGCCGCCTGGCGC
GAGCATTACGACCGGGCTGGCTACGCGCGGGTCGACCCGACGGTCAGTCACTGT
ACCCAGAGCGTACTGCCGATTTTCTGGGAACCGTCCATCTACCAGACGCGAAAG
CAGCACGAGTTCTTCGAGGAAGCCTCGGCCGCCGGCCTGGTGTATGGGCTGACC
ATGCCGCTGCATGGTGCTCGCGGCGAACTCGGCGCGCTGAGCCTCAGCGTGGAA
GCGGAAAACCGGGCCGAGGCCAACCGTTTCATAGAGTCGGTCCTGCCGACCCTG
TGGATGCTCAAGGACTACGCACTGCAAAGCGGTGCCGGACTGGCCTTCGAACAT
```

OK

Sequence Viewer

blue_green_oscillator — design 0 · 5082 bp · 20 slots

Slot 1 · igem-dist-tetR_Orthologs-pButR *(promoter, 46 bp)*
ttcGTGTCACTTTGACAGCAGTGTCACtcctaggtataatgctagc
Regenerate with Evo 2

Slot 2 · cello-Eco1C1G1T1-Q1 *(RBS, 41 bp)*
GTAAGCCATGCCATTGGCTTTTGATAGAGGATAACTACTAG
Regenerate with Evo 2

Slot 3 · BBa_C0179 *(CDS, 723 bp)*
ATGGCCTTGGTTGACGGTTTTCTTGAGCTGGAACGCTCAAGTGGAAAATTGGAGTGGAGCGCCATCCTCCAGAAGATGGCGAGCGACCTTGGATTCTCGAAGATCCTGTTCGGCCTGTTGCCTAAGGACAGCCAGGACTACGAGAACGCCTTCATCGTCGGCAACTACCCGGCCGCCTGGCGCGAGC
ATTACGACCGGGCTGGCTACGCGCGGGTCGACCCGACGGTCAGTCACTGTACCCAGAGCGTACTGCCGATTTTCTGGGAACCGTCCATCTACCAGACGCGAAAGCAGCACGAG… (723 total)
Regenerate with Evo 2

Slot 4 · BBa_B0010 *(terminator, 80 bp)*
CCAGGCATCAAATAAAACGAAAGGCTCAGTCGAAAGACTGGGCCTTTCGTTTTATCTGTTGTTTGTCGGTGAACGCTCTC
Regenerate with Evo 2

Export to GenBank | Close

presenter: Mike

Backup Demo

Enumerated Designs

Export All to CSV | Fetch Sequences

Assembled DNA sequences for 3 design(s) from
blue_green_oscillator
(12 unique parts, 1 batch MCP call)

>blue_green_oscillator design 0 | 5082 bp View

```
ttcGTGTCACTTTGACAGCAGTGTCACtcctaggtataatgctagcGTAAGCCA
TGCCATTGGCTTTTGATAGAGGATAACTACTAGATGGCCTTGGTTGACGGTTTT
CTTGAGCTGGAACGCTCAAGTGGAAAATTGGAGTGGAGCGCCATCCTCCAGAAG
ATGGCGAGCGACCTTGGATTCTCGAAGATCCTGTTCGGCCTGTTGCCTAAGGAC
AGCCAGGACTACGAGAACGCCTTCATCGTCGGCAACTACCCGGCCGCCTGGCGC
GAGCATTACGACCGGGCTGGCTACGCGCGGGTCGACCCGACGGTCAGTCACTGT
ACCCAGAGCGTACTGCCGATTTTCTGGGAACCGTCCATCTACCAGACGCGAAAG
CAGCACGAGTTCTTCGAGGAAGCCTCGGCCGCCGGCCTGGTGTATGGGCTGACC
ATGCCGCTGCATGGTGCTCGCGGCGAACTCGGCGCGCTGAGCCTCAGCGTGGAA
GCGGAAAACCGGGCCGAGGCCAACCGTTTCATAGAGTCGGTCCTGCCGACCCTG
TGGATGCTCAAGGACTACGCACTGCAAAGCGGTGCCGGACTGGCCTTCGAACAT
```

OK

Sequence Viewer

blue_green_oscillator — design 0 · 5082 bp · 20 slots · 1 edit · 5082 bp assembled

Slot 1 · igem-dist-tetR_Orthologs-pButR *(promoter, 46 bp)*

ttcGTGTCACTTTGACAGCAGTGTCACtcctaggtataatgctagc

Regenerate with Evo 2

Slot 2 · cello-Eco1C1G1T1-Q1 *(RBS, 41 bp)* (edited — Evo 2)

AAAGTAAAGAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

Regenerate with Evo 2

Evo 2 proposal — 41 bp · mean sampled prob 0.586 · 1724 ms · seed 46 bp 5' context

AAAGTAAAGAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

Original (41 bp):

GTAAGCCATGCCATTGGCTTTTGATAGAGGATAACTACTAG

Applied ✓ | Revert

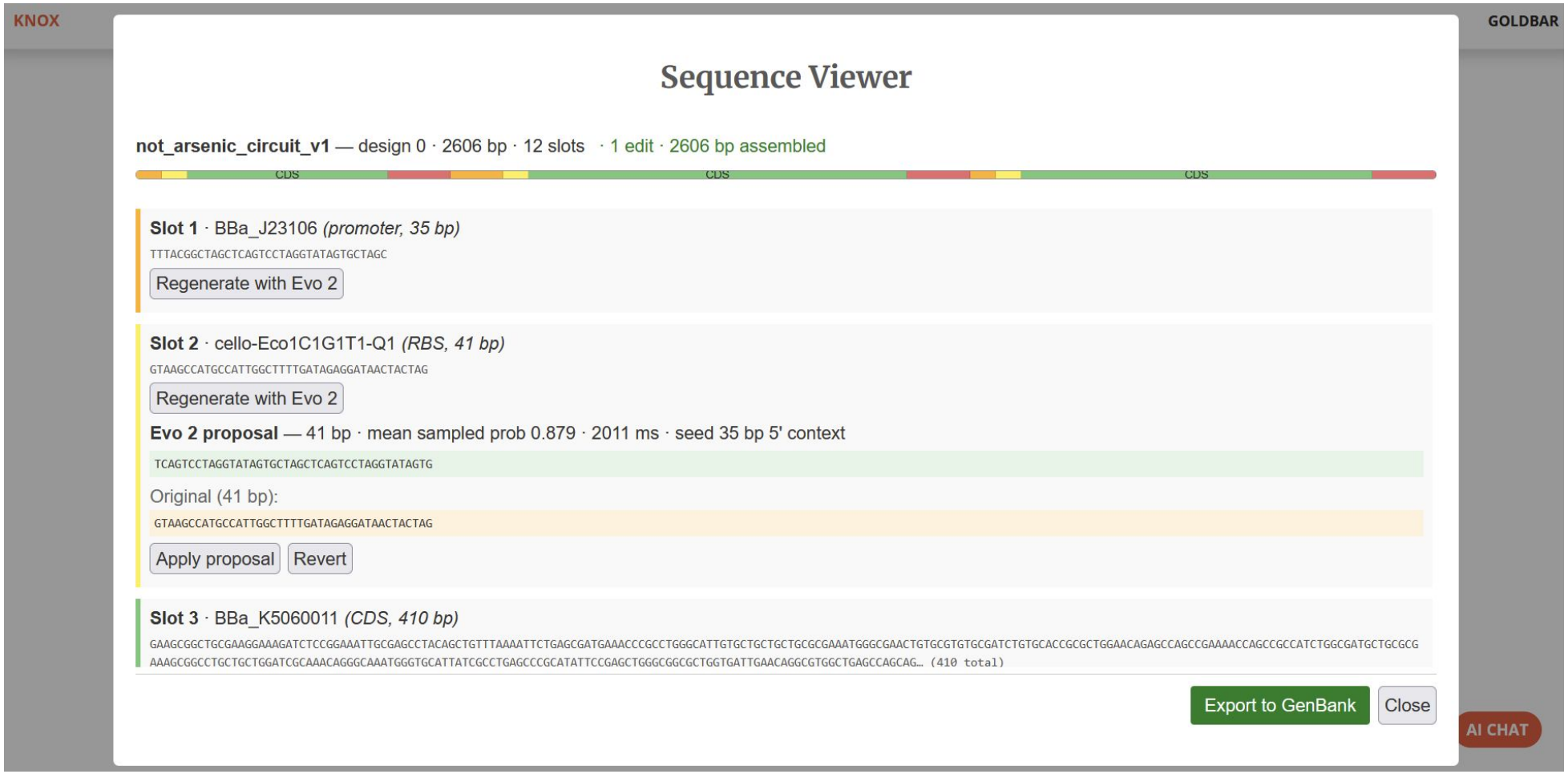
Slot 3 · BBa_C0179 *(CDS, 723 bp)*

ATGGCCTTGGTTGACGGTTTTCTTGAGCTGGAACGCTCAAGTGGAAAATTGGAGTGGAGCGCCATCCTCCAGAAGATGGCGAGCGACCTTGGATTCTCGAAGATCCTGTTCGGCCTGTTGCCTAAGGACAGCCAGGACTACGAGAACGCCTTCATCGTCGGCAACTACCCGGCCGCCTGGCGCGAGC
ATTACGACCGGGCTGGCTACGCGCGGGTCGACCCGACGGTCAGTCACTGTACCCAGAGCGTACTGCCGATTTTCTGGGAACCGTCCATCTACCAGACGCGAAAGCAGCACGAG… (723 total)

Export to GenBank | Close

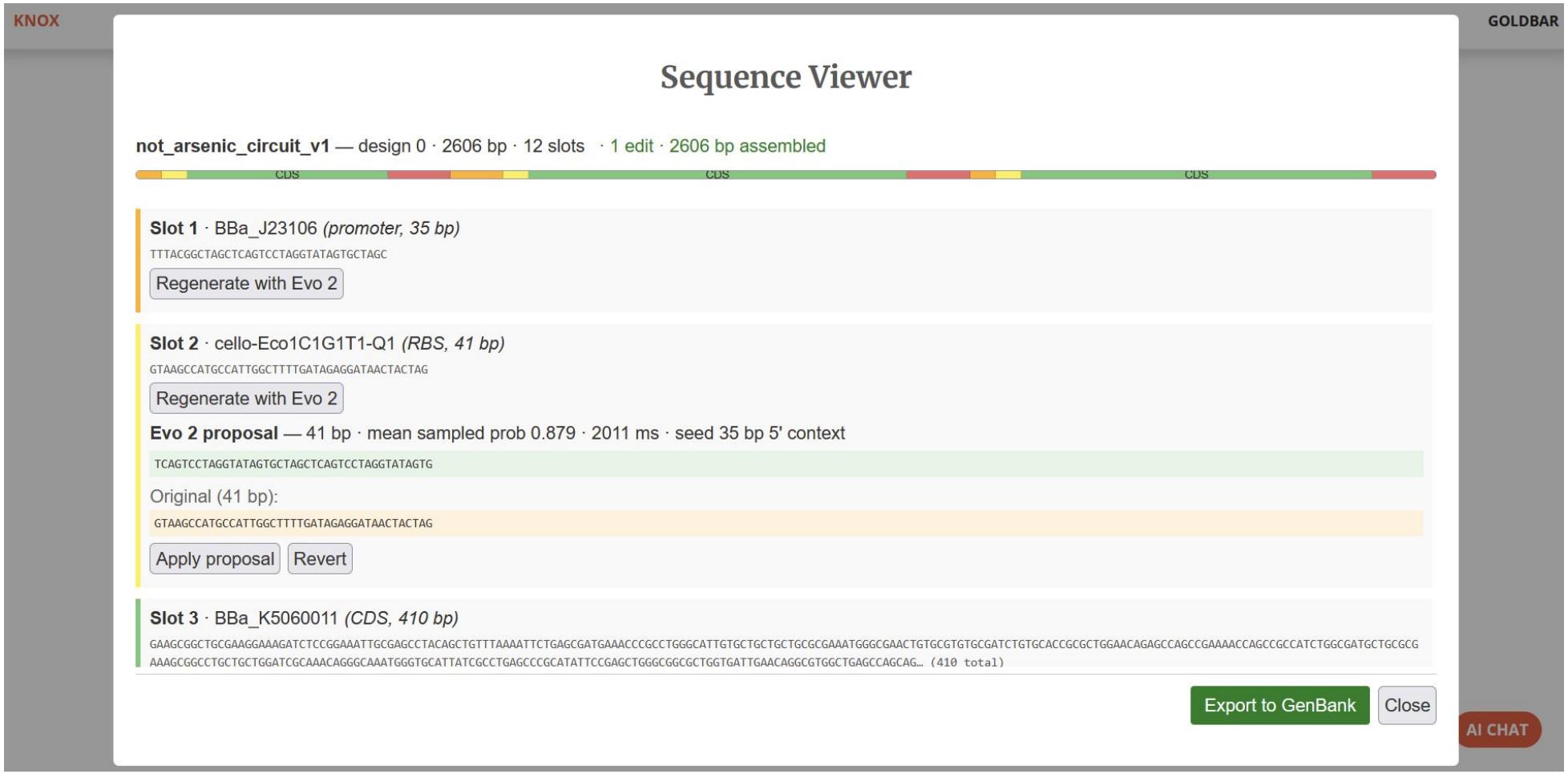
presenter: Mike

Backup Demo



presenter: Mike

Backup Demo



presenter: Mike